StoryCast: Research Document

Cherish’s Student project, October 2026

What I built

StoryCast is a small website with three pages where people can watch or listen to short stories. The idea behind it is simple: you shouldn't need to hear a story to get it. So every video has captions, and every story has the full transcript written out on the same page.
The three pages are the Home page (a short intro and the featured stories), the Story page (the video, the captions, the transcript and related stories) and the About page (why I made it and what accessibility features are in the site). Only one story is finished. The other two are marked "Coming soon".
I built it with plain HTML, Sass and a tiny bit of JavaScript. No frameworks.

Who it's for

I picked deaf and hard-of-hearing people as the main audience. I chose them because the problem is easy to test: if the sound is off, can you still get the whole story? That told me exactly what to build, which was captions and transcripts.
I also kept blind users and keyboard-only users in mind. Most of what helps them comes from writing the HTML properly: real headings, real buttons, landmarks like nav and main, and a focus outline you can see. It costs very little if you do it from the start.
Example user (made up for this project): Amina is a university student who is deaf. She watches stories on her phone on the bus with the sound off. She wants captions she can trust, and she likes to skim a transcript to find the part she cares about. She leaves quickly if a site plays sound by itself or the buttons are tiny.

What WCAG AA means

WCAG is the set of rules, written by the W3C, for making websites usable by people with disabilities. It has three levels: A, AA and AAA. AA is the one most schools, companies and governments ask for, so that is the one I aimed for. The rules are grouped under four ideas: content has to be perceivable, operable,

understandable and robust.

These are the rules that mattered most for my site:

Rule	In plain words	What I did
Captions (1.2.2)	Videos with speech need captions	Added a captions track (a .vtt file) to the video
Text version of audio and video (1.2.1, 1.2.3)	Give people the content as text too	Full transcript under the video
Contrast (1.4.3)	Text must stand out from its background	Calculated every colour pair. The lowest is 7.37:1 and AA needs 4.5:1
Keyboard (2.1.1)	Everything works without a mouse	Used real links and buttons
Bypass blocks (2.4.1)	Let people skip the menu	"Skip to main content" link on every page
Focus visible (2.4.7)	Show where you are when you press Tab	Thick amber outline on everything you can focus
Language of page (3.1.1)	Say what language the page is in	lang="en" on every page
Name, role, value (4.1.2)	Controls must make sense to a screen reader	Native elements first. ARIA only on the transcript button (aria-expanded) and the current page link (aria-current)

What other sites do

Before building, I looked at how a few big sites handle captions and transcripts. The notes below come from their own pages and from articles about them, listed in the sources at the end.

Site	What I found	What I took from it
TED	TEDx talks get transcripts made by volunteers, and those transcripts are also used to make translated subtitles	Treat the transcript as part of the page, not an extra
NPR	NPR says it aims to provide transcripts for its shows and podcasts, but they can appear after the audio because they get checked first. Many of its videos have captions	A transcript is worth having even when the story is audio only
BBC iPlayer	Some programmes come with subtitles, audio description and signed versions	Offer more than one way into the story where you can

Things I wanted to avoid:

- Players with controls you can't reach using the keyboard.
- Transcripts hidden behind several clicks.
- Low-contrast grey text.
- Sound or video that starts by itself.

How I planned the site

The structure is flat. Home links to the Story page and the About page, the Story and About pages link to each other, and every page has the same menu at the top, so Home is always one click away. The sitemap is a separate file in the docs folder.

For the look I wanted something calm that is easy on the eyes:

Element	Choice	Why
Background	Dark navy #0b1f3a	Calm, not glaring

Element	Choice	Why
Text	Off-white #f5f1e8	14.65:1 contrast on the background
Accent	Amber #ffb547	Links, buttons and focus outline. 9.40:1 on the background
Cards	#16315a	A bit lighter than the page so cards stand out. Amber on cards is 7.37:1
Font	Atkinson Hyperlegible	Made by the Braille Institute so letters are easy to tell apart

For layout I used CSS Grid for the card lists and the Story page, and Flexbox for the header and menu. The container query is on the story cards. Each card sits inside a wrapper box. When that box is wide, the photo goes beside the text, and when it is narrow the photo goes on top. So the card reacts to the space it has, not to the screen size.

The Sass is split into partials. The colours, fonts and spacing are variables in their own files, and the rest is split into base, layout and components. I used cascade layers (reset, base, layout, components) so the order of the styles is set on purpose, and BEM names like card, card__image and nav__link--current for the classes.

How I made it accessible

- Landmarks on every page: header, nav, main, footer, plus article, section and aside where they fit.
- One h1 per page, and headings that go in order.
- A skip link as the first thing you reach with Tab.
- Captions on the video and a full transcript that can be hidden or shown. The button uses aria-expanded so a screen reader knows if it is open.
- The current page in the menu is marked with aria-current.
- A figcaption that describes what you can see in the video.
- Empty alt text on the cover photos, because the story title right next to each one already says what

it is.

- Nothing plays by itself.
- The layout works on small screens.

How I tested it

Test	What I did	Result
Colour contrast	Calculated the ratio for each colour pair	Lowest is 7.37:1, so everything passes AA
Keyboard only	Tabbed through all 3 pages without the mouse	[add your result]
Lighthouse accessibility	Ran it in Chrome on each page	Home [score], Story [score], About [score]
Zoom	Browser zoom at 200%	[add your result]
Captions	Played the video with captions on	[add your result]

What I would do with more time

- Record the other two stories, including one that is audio only with its own transcript.
- Add audio description for the parts of the video that are only visual.
- Make the video file smaller so the page loads faster.
- Test with a screen reader and with real deaf and hard-of-hearing users. I only tested with tools and the keyboard, and that is not the same thing.

Sources

- Web Content Accessibility Guidelines (WCAG) 2.1, W3C
- Making Audio and Video Media Accessible, W3C WAI

- NPR Accessibility Statement
- Transcribe TEDx talks, TED
- Accessibility on BBC iPlayer, Ofcom
- Waterloo Road on BBC iPlayer with all accessibility options, Broadcast
- Atkinson Hyperlegible, Wikipedia
- The track element, MDN